

Design and Construction of a Foot Step Based Power Generation System Using Rack and Pinion Mechanism

Tanver Ahmed ¹, Ratul Anick Paul²

Department of Mechanical Engineering, Khulna University of Engineering & Technology, Khulna- 9203, BANGLADESH

ABSTRACT

Power generation through footsteps is a non-conventional energy that can be obtained while walking on certain arrangements like footpaths, stairs and platforms in crowded areas. This project represents the design and construction of a foot step based power generation system that uses rack and pinion mechanism which can convert potential energy from human body weight into mechanical energy and then electrical energy. A model is constructed on the basis of that design to demonstrate how it works. The performance of the device is measured. A model of 71.12 cm height, 45.72 cm width and 30.48 cm breadth having springs of spring constant 8.82 is able to take the load of a person of 80 kg. In this case, the model produces 863 rpm and generates 5.03 volt. As a small single unit, it's found to produce electrical power in little amount. If a setup is constructed with similar units connected together, it is possible to generate even sufficient power that can be used to satisfy local power demand.

Keywords: Footstep, Rack-pinion, Non-conventional Energy, Power Generation, DC Generator.

1. Introduction

Human beings are using energy at an increasing rate ever since they came on the earth a few million years ago. The extensive usage of energy has been resulted in an energy crisis over the few years. Therefore to overcome this problem, it's necessary to implement the techniques of optimal utilization of conventional sources for conservation of energy.

Recent technology seeks for cheap and clean ways to produce electricity. Due to a need to reduce carbon emissions and keep costs down, producing renewable energy is increasingly important for most of the governments and the technological industries at this moment. In our country, electricity is produced mostly from gas, coal and petroleum which are limited resources. Alternatives are required to find to satisfy the increasing demand as well as to reduce electricity consumption cost for the users. This project work proposes and shows an alternative way of power generation that can be used to satisfy the power requirement in small scale.

Walking is the most common activity in human life. When a person walks, he loses energy to the road surface in the form of inertia and thereby applies his weight on to the road surface, through footsteps on the ground during every step. This energy can be tapped and converted into the usable form such as in electrical form. This device, if embedded in the footpath, stairs, platforms etc. can produce electrical energy from the footsteps.

This device is designed and constructed in such a manner that when a man steps down his foot on it, the potential & kinetic energy both are transmitted to a shaft via rack and pinion mechanism, developing rotation in the shaft. This mechanical power can be used to rotate generator to produce electricity. This power generation

process can be considered as a non-conventional and pollution.

2. Working Principle

The infrastructure of the whole system is to be made of metal box channel. It's known to us that helical spring can absorb any shock or load being compressed. When any human walks over the pressure pad at the top most of the structure then the spring will get compressed at its solid height. Because of this downward direction of force, rack will settle down which is meshed along with pinion. Then the linear motion of rack will be converted into rotary motion & a rotation will be obtain in the power transmitting shaft which is guided with bearings at both of the ends.

Bearing is used to hold the power transmitting shaft & to ensure the relentless rotation without friction. When the pressure pad will get released from applying load then the spring will retrieve the pressure pad to its previous height, then the rotation will be reverse. If the rotation is clockwise during applying load, so the rotation will be anti-clockwise during releasing load. But it's required to obtain unidirectional rotation that's why a chain & sprocket mechanism will be used to obtain the unidirectional rotation.

A flywheel will also be used to supply continuous rotation using its inertia of rotation. Finally this mechanical rotation will be utilized to generate the electricity. For electricity generation a DC generator will be used to generate electricity utilizing mechanical rotation as generator needs a rotation of armature to cut the magnetic flux. After this the induced electromotive force is converted into electricity. This electricity can be stored into battery. In short it can be summarized that a non-conventional energy designated as human foot step is utilized to produce electricity.

* Corresponding author. Tel.: +88-01912934563
E-mail address: ratulanick@gmail.com

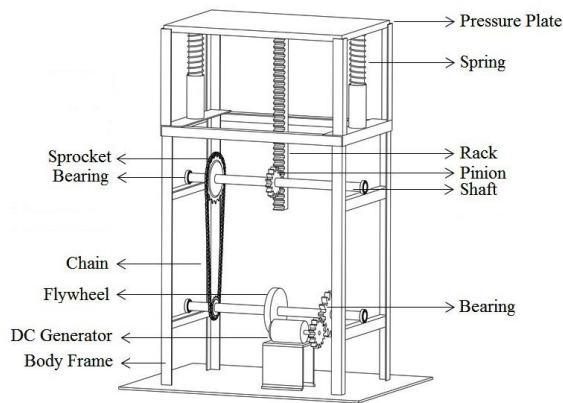


Figure 2.1 Isometric View of the Power Generation System

3. Design Criterion of Required Components

3.1 Design of Spring

The calculation of spring design is as bellow.

Some values have been assumed approximately in order to continue the following calculation. The spring has been designed for load limited to 80 kg. Hence two springs have been used. 40 kg load has been considered to be acted on each spring individually. Assuming light service and material to be mild steel equation of design stress on springs, from Table AT-17 [7]

$$S_{ds} = 0.405 \times \frac{182}{D_w^{0.1}} = \frac{73.71}{D_w^{0.1}} \text{ ksi}$$

Assuming, Wahl factor [7], $K = 1.15$ and mean spring diameter, $D_m = 3.8 \text{ cm} \approx 1.5 \text{ inch}$

$$F = 40 \times 2.2 = 88 \times 10^{-3} \text{ kips}$$

Induced stress,

$$S_s = \frac{K \times 8 \times F \times D_m}{\pi \times D_w^3} = \frac{1.15 \times 8 \times 88 \times 10^{-3} \times 1.5}{\pi \times D_w^3} = \frac{0.39}{D_w^3} \text{ ksi}$$

$$\text{For design condition, } S_s = S_{ds} \\ \Rightarrow \frac{0.39}{D_w^3} = \frac{73.71}{D_w^{0.1}} \therefore$$

$$D_w = 0.17 \text{ inch}$$

$$= 0.43 \text{ cm}$$

Spring Index,

$$C = \frac{\text{Mean Spring Diameter}}{\text{Wire Diameter of Spring}} = \frac{1.5}{0.17} = 8.82$$

Wahl factor,

$$K = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = \frac{4 \times 8.82 - 1}{4 \times 8.82 - 4} + \frac{0.615}{8.82} = 1.16$$

Now, design stress and induced stress are calculated separately for checking.

Design stress,

$$S_s = \frac{1.16 \times 8 \times 88 \times 10^{-3} \times 1.5}{\pi \times 0.17^3} = 79.36 \text{ ksi}$$

It is found that design stress is more than induced stress. So, the design is okay.

The values obtained from the calculation are as followed:

Material of spring is mild steel.

Maximum load = 40 kg = 392.4 N
Wire diameter of spring = 0.43 cm
Mean diameter of spring = 3.8 cm
Spring index = 8.82

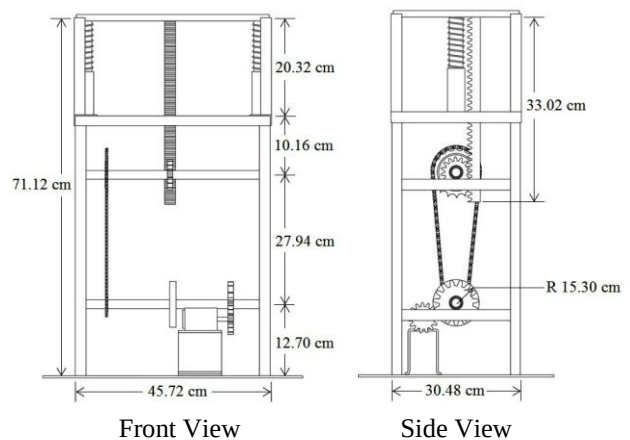


Figure 3.1 Drawing of the System with Dimensions

3.2 Design Assumption of Spring

Since the springs of exact calculated measurements are not available in market and it is not possible to manufacture those in our labs, the springs having nearest specification have been managed from local market. The selected spring specification are as:

Material is mild steel.

Wire diameter of spring = 0.45 cm

Mean diameter of spring = 3.875 cm

Spring index = 8.61

Free length = 20.32 cm

Solid height = 5.25 cm

Maximum Deflection, $\delta = 10.16 \text{ cm}$

3.3 Design Assumption of Power Transmitting Elements

3.3.1 Rack and Pinion

Material is cast iron.

Length of rack = 12 inch $\approx 30.48 \text{ cm}$

Width of rack = 1.9 cm

No. of teeth on rack = 21

Diameter of addendum circle of pinion = 5.5 cm

Diameter of dedendum circle of pinion = 4.54 cm

No. of teeth of pinion = 21

Width of teeth = 1 cm

3.3.2 Chain Sprocket and Freewheel

No. of teeth of sprocket = 48

No. of teeth of freewheel = 18

3.3.3 Spur Gear

Diameter of addendum circle of gear = 14.46 cm

Diameter of dedendum circle of gear = 13.5 cm

No. of teeth of Gear = 56

Width of teeth = 1.3 cm

4. Construction of the System

4.1 Mechanical Setup

The whole system consists of metal frame infrastructure, 2 pieces of compression spring, 4 pieces of ball bearing, 2 pieces of shaft, rack & pinion, chain-sprocket & freewheel, gear & pinion, flywheel, DC generator. The total setup was done such a way that the load was to be applied on the system so that the system could generate the electricity. The required components were used on the basis of approximation and availability.

The CAD design schemed in Solidworks software shows the model view of the system in Figure 3.1. On the other hand, photo captured view of the system after construction is represented in Figure 3.2 as following.



Figure 4.1 CAD Design of the System

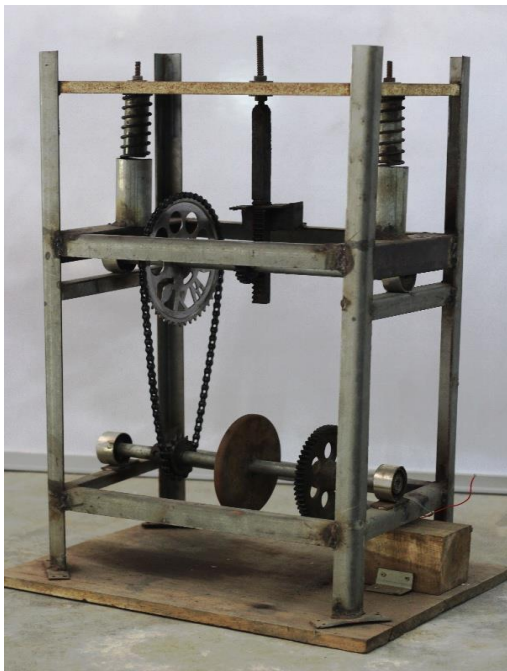


Figure 4.2 Photographic View

4.2 Electronic Setup

The main concern of our system is to generate electricity. The indication of the power output from the system is shown by illuminating LED lights. But due to the fluctuation of the shaft speed, the lights dim and get bright successively. For continuous output, an electronic setup was attached with the system which consisted of diode, capacitor, bread board, wire etc.

Capacitors are connected in parallel with the DC power circuits of most electronic devices to smooth current fluctuations for signal or control circuits. A capacitor can store electric energy when it is connected to its charging circuit. And when it is disconnected from its charging circuit, it can dissipate that stored energy, so it can be used like a temporary battery. Capacitors are commonly used in electronic devices to maintain power supply while batteries are being changed. A capacitor having capacity of 2200 μF was used in the system.

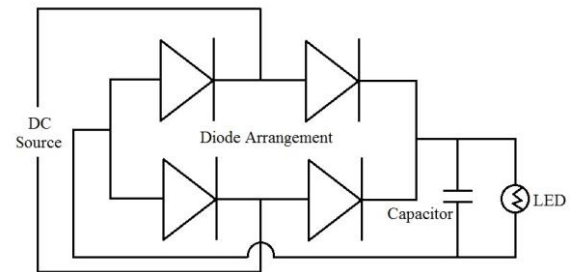


Figure 4.3 Circuit Diagram of Electronic Setup

Diode is the device which can pass the current in one direction only and the power supply only applies the positive voltage to the circuit. The diode's arrow points in the direction of forward current flow. The number of diodes used is 1N4007.

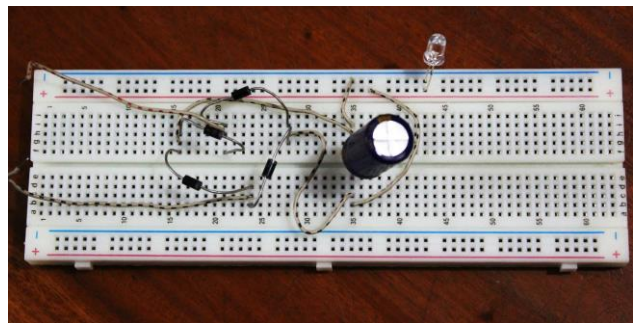


Figure 4.4 Photographic View of the Electronic Circuit Setup

5. Experimental Data Collection

The system has been constructed for generating electricity using human footstep. The main concern of the system is human body weight which was considered as load acting on the system. The loads were considered to carry on the performance testing such as 40 kg, 50 kg, 60 kg, 70 kg and 80 kg approximately.

Table 5.1: Data table for primary shaft speed against various loads

Load (kg)	Primary Shaft Speed (RPM)										Average Speed (RPM)
	Observation No.										
	1	2	3	4	5	6	7	8	9	10	
40	254	249	243	231	257	261	247	252	243	231	246.80
50	273	282	269	253	244	280	264	271	252	248	263.60
60	287	297	319	316	275	269	287	279	266	259	284.50
70	299	287	310	322	289	301	318	298	315	285	302.40
80	342	333	298	328	308	299	316	338	329	335	322.60

Sample Calculation for Theoretical Speed of Driven Shaft:

Pitch diameter of the driver gear, $D_g = 14.21$ cm

Pitch diameter of the driven pinion, $D_p = 5.25$ cm

Hence, the speed of the driven shaft, $N_p = \frac{D_g}{D_p} \times$ the

speed of the primary shaft, $N_g = \frac{14.21}{5.25} \times 246.80$
 $= 668.83$ rpm. [8]

Table 5.2: Data table for driven shaft speed against various loads

Load (kg)	Primary Shaft Speed, N_g (RPM)	Theoretical Speed of Driven Shaft, N_p (RPM)	Actual Speed of Driven Shaft (RPM)
40	246.80	668.83	661.00
50	263.60	714.36	702.00
60	284.50	771.00	759.00
70	302.40	819.50	809.00
80	322.60	874.25	863.00

Table 5.3: Data table for spring deflection against various loads

Load (kg)	Initial Height of Spring (cm)	Height Under Load (cm)	Deflection (cm)
40	20.32	16.46	3.86
50	20.32	15.37	4.95
60	20.32	14.07	6.25
70	20.32	12.95	7.37
80	20.32	11.56	8.76

Table 5.4: Data table for voltage against various loads

Load (kg)	Voltage (Volts)										Average Voltage (Volt)
	Observation No.										
	1	2	3	4	5	6	7	8	9	10	
40	3.41	3.39	2.97	3.62	3.35	2.87	3.40	3.53	2.92	3.59	3.31
50	3.97	4.12	3.55	3.79	4.05	3.40	3.83	3.55	3.81	4.11	3.82
60	4.51	3.97	4.21	4.53	3.83	4.78	4.61	3.55	4.89	4.32	4.32
70	4.88	4.96	5.05	4.67	4.58	4.80	4.41	5.03	4.91	5.11	4.84
80	5.12	4.93	4.81	5.09	5.15	4.87	4.96	5.07	4.94	5.17	5.03

Sample Calculation for Power Produced in Generator Shaft

From Table 4.2 and 4.3,

Total load acting on pressure plate, $W = 40 \times 9.81 = 392.4 \text{ N}$

Load on each spring, $L = \frac{392.40}{2} = 196.2 \text{ N}$

Deflection of spring for 40 kg load, $\delta = 3.86 \text{ cm}$

Speed of the driven shaft for 40 kg load, $N = 661 \text{ rpm}$

So, Spring Constant, $k = \frac{L}{\delta} = \frac{196.2}{3.86} = 50.80 \text{ N/cm}$

Now, while applying the load on the pressure plate, the springs absorb a certain amount of energy. When the load is removed, then the springs create an upward force to mitigate the downward force and send back to its previous position. So, in order to find out the average value of acting force, the force was calculated at different points within the deflection.

Downward force acting at initial position, $F_1 = 392.40 \text{ N}$

Downward force acting at 1.29 cm, $F_2 = 392.40 - (50.80 \times 1.29 \times 2) = 260.94 \text{ N}$

Downward force acting at 2.58 cm, $F_3 = 392.40 - (50.80 \times 2.58 \times 2) = 129.87 \text{ N}$

Downward force acting at 3.86 cm, $F_4 = 392.40 - (50.80 \times 3.86 \times 2) = 0.22 \text{ N}$

Hence, average downward force, $F = (F_1 + F_2 + F_3 + F_4)/4 = 195.86 \text{ N}$

Torque produced in the main shaft, $T = F \times R_{\text{pinion}} = 195.86 \times 2.75 = 538.62 \text{ N-cm} = 5.39 \text{ N-m}$

Now, the torque of the main shaft is transmitted to primary shaft which both are connected by chain sprocket. The teeth number of sprocket and freewheel are 48 and 18 respectively.

Torque transmitted to primary shaft = $\frac{\text{Teeth No. of Freewheel}}{\text{Teeth No. of Sprocket}} \times \text{torque in the main shaft}$
 $= \frac{18}{48} \times 5.39 = 2.02 \text{ N-m}$

Again, the torque of the primary shaft is transmitted to generator shaft which both are connected by gear and pinion.

The teeth number of gear and pinion are 48 and 18 respectively.

Torque transmitted to generator shaft =

$\frac{\text{Teeth No. of Pinion}}{\text{Teeth No. of Gear}} \times \text{torque in the primary shaft} = \frac{21}{56} \times 2.02 = 0.76 \text{ N-m}$

Power in the generator shaft (driven shaft), $P = \frac{2\pi NT}{60}$
 $= \frac{2 \times \pi \times 661 \times 0.76}{60} = 52.61 \text{ Watt [8]}$

6. Results

The power output for each load in the driven shaft was calculated individually following the calculation in the previous chapter. The calculated power outputs are given below.

Power output for 40 kg load, $P_1 = 52.61 \text{ Watt}$

Power output for 50 kg load, $P_2 = 85.28 \text{ Watt}$

Power output for 60 kg load, $P_3 = 124.79 \text{ Watt}$

Power output for 70 kg load, $P_4 = 166.04 \text{ Watt}$

Power output for 80 kg load, $P_5 = 214.18 \text{ Watt}$

A load applied (kg) versus power output (watt) curve has been plotted in the following graph using these data.

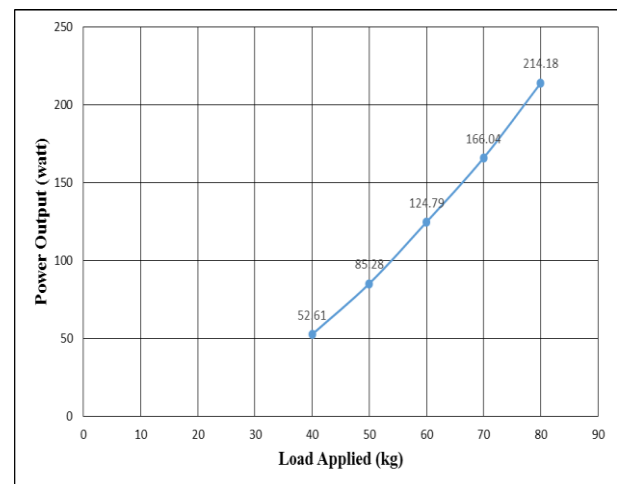


Figure 6.1 Power Output (W) versus Load Applied (kg) Curve

From Table 5.4 load applied (kg) versus voltage (volt) curve has been plotted in the following graph using obtained data.

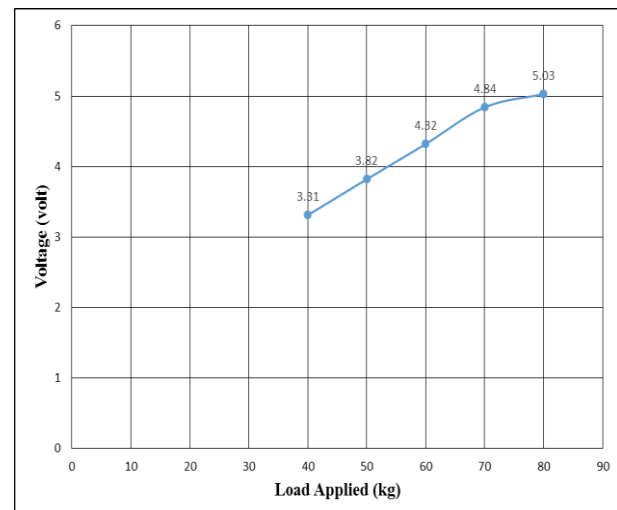


Figure 6.2 Voltage (volt) versus Load Applied (kg) Curve

From Table 5.2 load applied (kg) versus theoretical speed of the driven shaft and actual speed of the driven shaft curve has been plotted in the following graph as well as both curves are compared together.

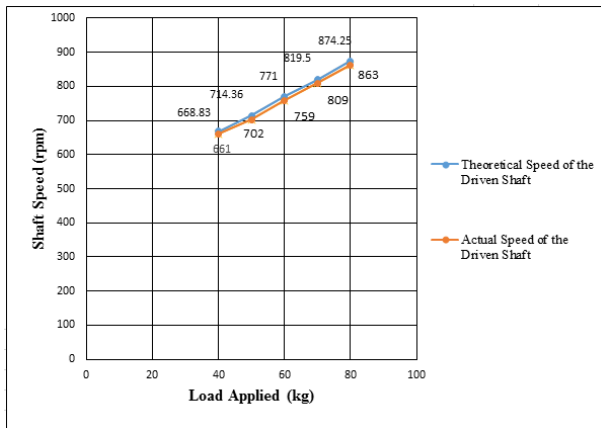


Figure 6.3 Shaft Speed (rpm) versus Load Applied (kg) Curve

7. Discussion

The purpose of the project was to produce electricity using human footsteps. In the project, the system was designed solving design calculation as well as taking some assumptions. Due to the unavailability of the materials and equipment in the market, most of the materials were selected nearest to the calculated values. The whole system has been constructed in mechanical workshop according to design. The system was design and constructed concerning of human being weighed of 80 kg maximum. The performance was tested to differ the actual values to the theoretical values. All the data were taken for the load of human body weight from 40 kg to 80 kg. The speed of the shafts used in the system was recorded with a reflector tachometer for each load applying on the system. The speed was fluctuating due to the variation of the shaft speeds. So, the values of shaft speeds were taken for 10 times and then average value was considered. It consumed much time to stable the value of shaft speed in tachometer. The spring deflection was also calculated for each load applying on the system. For generating electrical power a DC generator of 12 volts was used connecting to the primary shaft with gear set. The voltage of generated electricity was recorded for each load applying on the system. The values were taken for 10 times and then the average value was considered. The gear ratios of rack & pinion, sprocket & freewheel and gear set were maintained according to the calculation of speed and torque. The power output was calculated using torque and speed theoretically. The load applied versus power output curve, load applied versus voltage cure and load applied versus shaft speed curve were plotted in graphs using the recorded values. The theoretical and actual values were compared from the graphs.

8. Conclusion

In the project, an idea of non-conventional energy source has been represented as well as a model of the system based on the idea has been constructed. Performance test of the model has been done. The model satisfied the design criteria. 863 rpm with 5.03 voltage was generated in the maximum design condition for a load of 80 kg.

NOMENCLATURE

S_{ds}	: Design Stress, ksi
S_s	: Induced Stress, ksi
D_w	: Wire Diameter of Spring, cm
D_m	: Mean Spring Diameter, cm
F	: Force, N
C	: Spring Index
K	: Wahl Factor
D_g	: Gear Diameter, cm
D_p	: Pinion Diameter, cm
N_g	: Speed of the Primary Shaft, rpm
N_p	: Speed of the Driven Shaft, rpm
L	: Load, N
W	: Total Load, N
δ	: Spring Deflection, cm
K	: Spring Constant
T	: Torque, N-cm
N	: Speed, rpm
P	: Power, Watt

REFERENCES

- [1] <http://www.scienceclarified.com/scitech/Energy-Alternatives/The-Development-of-Energy.html>
- [2] <http://www.economicsdiscussion.net/notes/conventional-and-non-conventional-sources-of-energy/2177>
- [3] http://www.ucsusa.org/clean_energy/our-energy-choices/renewable-energy/environmental-impacts-of.html
- [4] <http://www.sorbothane.com/kinetic-energy.aspx>
- [5] <http://www.preservearticles.com/201104235869/what-are-the-non-conventional-sources-of-energy.html>
- [6] Brooks, G. A., Fahey, T. D. & Baldwin, K. M. (2005). Exercise Physiology: Human Bioenergetics and Its Applications. McGraw-Hill, Boston.
- [7] Virgil Morning Faires, "Design of Machine Elements", Fourth Edition, Macmillan Company, New Work.
- [8] V B Bhandari, "Design of Machine Elements", Third Edition, Tata McGraw Hill Education Private Limited, New Delhi.
- [9] B.L Theraja, "Fundamentals of Electrical Engineering and Electronics", 2006 Edition, S.Chand Publication, New Delhi.
- [10] <http://www.electricaleasy.com/2014/03/flemings-left-and-right-hand-rule.html>